

BITUNG, Indonesia

WMO: 970160

Lat: 1.4431N

Lon: 125.1800E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	74.0	74.8	67.2	99.9	81.5	68.5	104.7	81.8	17.5	83.0	14.4	83.0	0.4	160	0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.6	92.1	78.2	91.4	78.3	90.7	78.3	80.8	88.6	80.1	88.0	79.9	87.7	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	147.7	85.4	78.0	145.5	85.1	77.4	142.7	84.7	44.4	88.9	43.9	88.6	43.5	88.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	11.5	9.7	70.5	95.6	2.7	2.9	68.5	97.7	66.9	99.5	65.4	101.1	63.4	103.2		
(5)			67.3	82.8	2.2	1.3	65.8	83.8	64.5	84.5	63.2	85.3	61.6	86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.7	83.3	82.8	83.0	82.6	82.7	82.2	81.8	82.0	82.3	82.7	82.9	83.5		
	DBStd	1.63	1.76	1.76	1.60	1.61	1.58	1.78	1.45	1.43	1.38	1.41	1.27	1.53		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11923	1033	920	1024	978	1014	966	987	991	970	1014	988	1038		
	CDD65	6448	568	500	559	528	549	516	522	526	520	549	538	573		
	CDH74	72142	6513	5594	6314	5906	6266	5674	5577	5613	5754	6209	6056	6666		
	CDH80	25607	2501	2107	2439	2193	2258	1814	1566	1559	1855	2286	2364	2665		
(14) Wind	WSAvg	3.1	3.6	3.3	2.8	2.4	2.4	3.0	3.7	4.6	3.7	2.6	2.0	2.8		
(15) Precipitation	PrecAvg	99.1	12.7	10.4	9.6	8.4	9.1	8.1	5.2	3.7	3.7	6.7	10.4	10.6		
	PrecMax	141.4	24.1	21.6	17.4	14.1	13.5	14.3	13.1	11.6	9.3	13.3	18.0	15.6		
	PrecMin	61.1	4.3	2.2	1.3	1.9	3.7	0.0	0.0	0.0	0.1	0.7	6.2	6.2		
	PrecStd	18.6	4.6	4.5	4.6	3.1	2.5	3.2	3.5	3.2	2.6	3.3	2.9	2.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	92.5	92.8	92.3	90.9	90.3	89.3	89.4	90.1	91.3	92.3	92.7		
		MCWB	78.1	76.9	77.3	78.3	79.6	79.6	78.9	78.7	78.7	79.0	79.2	78.8		
	2%	DB	91.3	91.2	91.5	90.8	89.9	89.1	88.1	88.0	88.9	90.0	91.0	91.7		
		MCWB	78.1	77.4	77.3	78.6	79.4	79.1	78.4	78.2	78.4	78.8	79.0	78.7		
	5%	DB	90.2	90.0	90.6	89.6	88.9	88.0	87.0	87.0	87.9	89.2	90.0	90.6		
		MCWB	78.0	77.4	77.4	78.9	79.2	78.7	78.1	77.8	78.1	78.7	78.9	78.6		
	10%	DB	89.0	88.8	89.3	88.4	87.8	86.8	86.0	85.9	86.9	88.1	89.0	89.4		
		MCWB	77.8	77.3	77.5	78.7	79.0	78.3	77.6	77.4	77.7	78.4	78.8	78.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.8	80.6	80.1	81.8	81.0	80.5	80.0	79.8	80.0	80.7	80.9	80.9		
		MCDB	89.3	88.5	88.6	88.2	88.5	88.4	87.3	87.0	88.0	88.7	89.3	89.6		
	2%	WB	79.8	79.4	79.3	80.6	80.2	79.8	79.2	79.0	79.2	79.9	80.0	80.1		
		MCDB	88.4	87.7	87.8	87.7	87.6	87.3	86.3	86.2	86.8	87.8	88.4	88.8		
	5%	WB	79.0	78.7	78.8	79.8	79.8	79.2	78.6	78.3	78.8	79.3	79.6	79.5		
		MCDB	87.7	87.2	87.3	87.0	87.1	86.2	85.6	85.3	86.2	87.0	87.9	88.1		
	10%	WB	78.4	78.1	78.2	79.2	79.3	78.8	78.1	77.9	78.2	78.9	79.1	79.0		
		MCDB	86.6	86.1	86.3	86.3	86.3	85.5	84.7	84.5	85.3	86.5	87.4	87.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.2	12.8	12.3	11.0	9.5	8.7	8.5	10.0	12.0	13.4	12.6		
		MCDBR	13.4	13.7	14.6	13.8	12.2	11.0	10.1	9.7	10.9	13.0	14.5	14.1		
	5% WB	MCWBR	4.9	4.8	4.9	5.5	5.0	4.6	4.6	4.5	5.0	5.5	5.6	5.1		
		MCWBR	12.9	12.9	13.2	12.7	11.6	10.3	9.7	9.1	10.4	12.3	13.8	13.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.419	0.424	0.428	0.428	0.423	0.420	0.418	0.427	0.438	0.479	0.442	0.426		
		taud	2.489	2.464	2.452	2.443	2.465	2.485	2.498	2.456	2.410	2.282	2.395	2.459		
	Ebn at Noon	Ebn at Noon	288	288	285	279	273	270	272	275	278	269	279	285		
		Edh at Noon	36	37	38	37	35	34	34	36	39	44	39	36		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1586	1715	1818	1796	1728	1595	1692	1828	1936	1875	1739	1632		
	RadStd	RadStd	91	143	146	128	108	104	149	110	125	110	101	115		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.54	N/A	N/A	+1.69	+1.13	+0.81	N/A	N/A	N/A	N/A

Nomenclature: See separate page